

Review

Review on alternatives for the reduction of textile microfibers emission to water

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ABSTRACT

The microplastics (MPs) are considered one of the most threatening pollutants. One of the main concerns is their continuous and cumulative flow to water environments, as they are very difficult to be removed. Microfibers (MFs) are a significant type of MPs, with textile articles as one of the most renowned sources. This review aims to provide the current status of these MFs as pollutants, discussing possible alternatives from the manufacturing until the final disposition of MFs. There are many alternatives to reduce these pollutants from reaching the environment but also gaps that need to be further evaluated and addressed. Besides, it should be noticed that alternatives could be complementary between them. Some viable and non-contaminating solutions to reduce this pollution are currently on the market. Also, one relevant aspect is the final disposition or usage of the retained MFs to avoid them from reaching aquatic environments.

1. Introduction

Water scarcity is becoming an increasing problem and the survey of its quality is a major concern. For this reason, numerous studies have been focused on water purification (Elwakeel et al., 2020; He et al., 2022) and wastewater remediation (Ahmed M Elgarahy et al., 2021a, 2021b; C. Zhou et al., 2022). Special attention must be paid to the removal of non-biodegradable pollutants that can provoke serious threats due they can be accumulated in the environment and introduced in the trophic chain (Ahmed M Elgarahy et al., 2021a, 2021b). There is a huge amount of non-biodegradable pollutants, such as antibiotics, dyes, microplastics (MPs) and microfibres (MFs).

The still under-revision definition of the European Chemical Agency (ECHA) indicates that the microfibers (MFs) are particles with a length to diameter ratio >3 and a maximum length of 15 mm (ECHA, 2019). If those come from chemically modified and/or non-biodegradable polymers, they are considered as a type of microplastics (MPs). These pollutants have been constantly found contaminating every ecosystem. It has been estimated that there are from 15 to 51 trillion floating MPs in marine environments and 14 million tons in the top 9 cm of sediments of the world's oceans (Barrett et al., 2020; UNEP and GRID-Arendal, 2016; van Sebille et al., 2015). MPs can be divided into primary or secondary. Primary are those emitted into the environment in a MPs size range (<5

mm) where as secondary are generated in the environment from physical degradation and fragmentation processes of larger plastic debris (Ahmed M Elgarahy et al., 2021a, 2021b). In this way, primary MPs include a wide variety of sources (e.g., MFs detached from textile garments, plastic pellets, tire dust); while secondary MPs have their origin in mismanaged plastic garbage (Boucher and Friot, 2017). Textile MFs have received significant attention as these have been extensively found across the environment. For instance, Alavian Petroody et al. (2020) reported that most of the MPs found in water treatment plants were MFs, from which polyester MFs were identified at the highest concentration, followed by polyamide and acrylic. This is in line with the main synthetic fibers that are manufactured across the world.

Regarding their impacts, the ingestion across the trophic chain has extensively been reported, as these particles have been found in at least 200 species (Collignon et al., 2012; Fossi et al., 2014; GESAMP, 2015a; Ohkubo et al., 2020; Patterson et al., 2019). Also, some works have found that these particles can potentially have negative effects, such as MPs' retention, trophic transfer, and endocrine disruption, among others (Jemec et al., 2016; Nelms et al., 2018; Rochman et al., 2014; Welden and Cowie, 2016). Moreover, MPs can behave as long-distance vectors for invasive species and hydrophobic compounds (Browne et al., 2013; Mao et al., 2020; Rochman et al., 2013; Ta and Babel, 2020; Turner et al., 2020). Besides, these contaminants have been identified in

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¹ In the memory of our beloved Francisco Belzagui. INTEXTER will always remember you.

human-consumption products as seafood, tap and bottled water, fruits, vegetables, and table salt (Abidli et al., 2019; G. Chen et al., 2021; Cox et al., 2019; Oßmann, 2021; Rochman et al., 2015; Schymanski et al., 2018; Shruti et al., 2020; Teng et al., 2019; Van Cauwenbergh and Janssen, 2014; Yang et al., 2015). Nevertheless, the potential risks for human health are still an unknown area of study (Vital et al., 2021).

This review aims to evaluate the different alternatives suggested to reduce the textile MFs' pollution. The review is divided into four levels: (1) textile manufacturing, (2) garments laundering, (3) used water treatment plants, and (4) gaps that must be treated to close the loop of this specific MFs contamination. Fig. 1 shows a summary of the information included in this document. This figure also illustrates that there is still a lot of gaps to find a circular trend for MFs.

To be more explicit, (1) textile manufacturing refers to all treatments and proceedings applied in the production of textile articles in general before their use. (2) Garments laundering involves the washing and drying processes. (3) The used water treatment plants section includes what happens when the MFs contained in the effluents from laundering processes reach a possible water treatment plant and the effects that these particles have on the treatments. And (4), The gaps of this contamination, indicates criteria, requirements and alternatives to reduce this pollution. The MFs that can be generated in the daily use or final disposal of a textile article are out of the scope of this study.

2. Manufacturing, dyeing, and industrial processes

The textile or apparel industry is the business for the manufacture of fibers, yarns, fabrics, clothing, and articles for home and/or decoration. Textile and clothing activities present different treatments which have enormous variations between them (Bullon et al., 2017). In this sense, there is still a lack of rigor regarding MFs' reduction techniques applied in the manufacturing process of the textile industry. Besides, there are uncountable textile mills spread around the globe. Moreover, developing countries produce 50% of the world's textile exports and 75% of the world's clothing exports. In some of these countries, environmental regulations are not usually a governmental priority and water treatment plants are scarce (Mara, 2004) (UN-Water, 2021).

Textiles release MFs during production, use, and at end-of-life disposal. For this reason, this industry can play an important role in the reduction of MFs by upgrading their processes and their products (Ellen MacArthur Foundation, 2017; Henry et al., 2019). Also, according to Xu et al. (2021), the MFs should be included in the concept of the circular economy, providing a regulatory framework to address this pollution (Xu et al., 2021). However, the textiles and apparel industry accounts for a very high percentage of total manufacturing occupations in many countries where poverty mitigation is a central issue (InfoDev,

2008). This complicates the challenging task to achieve a global adaptation to sustainable techniques that target the reduction of MFs' detachment. Besides, the term "microfiber" is not usually found in books or references for improvements in the sustainability in the textile industry (e.g., Muthu (2017)). This is an indirect demonstration that textile MFs are not receiving the attention that they should. The trend is more focused onto the customer expectation fulfillment or the currently "fast-fashion" trend and the waste that it generates. This last topic is another problem that must be treated as more than 70% of the annual fiber production for clothing is burnt or disposed in landfills, which involves a great wastage of every kind of resources (water, energy, textiles, etc.) (Rese et al., 2022) Rahman et al. (2022).

There are many examples of improved techniques to enhance productivity. For instance, in recent decades several new spinning systems have been introduced to the industry. These systems have resulted in important enhancements in yarn productivity but not necessarily in yarn quality (Islam, 2019; Nergis, 2017). Yet, there are some studies that are looking forward sustainable techniques like using wet spinning of fungus to create monofilament yarns (Svensson et al., 2021). The type of fabric (woven, knitted, etc.) also seems to play an important role. For example, De Falco et al. (2018b) reported that woven polyester released the highest number of MFs when compared to knitted polyester and woven polypropylene. Carney Almroth et al. (2018) observed that polyester fleece fabrics shed much more MFs than knitted fabrics made of polyester, acrylic, and polyamide. They also reported that high twisted yarns are less prone to detach MFs, which is in line with the observation exposed in Table 1.

On the other hand, in a study conducted by Zambrano et al. (2020), an evaluation of cotton knitted fabrics was done. They reported relevant information that can be furtherly assessed with synthetic fabrics. They found that the treatments applied during textile processing influenced the MFs released during laundering. In this sense, fabrics treated with softeners generated the longest MFs, while durable press and water repellent generated the shortest ones. They pointed out that, in general, fabrics with more abrasion resistance, higher friction coefficient, and less softness (i.e., fuzz or hairiness) reduce the detachment of MFs. In this line, a Life European project evaluated textile procedures and published a guideline of "better practices" and main barriers for any industry relevantly linked to the textile chain (Mermaids, 2018). The key issues identified were the fiber (fineness, irregularities, length), the yarn (number of plies, twist, count), the fabric (structure, density, processes as dyeing and finishing), and garment washing in factories and their wastewater management. A summary of their observations is listed in Table 1.

Textile parks also are a source of MFs. For instance, the work of Zhou et al. (2020) explored the MFs' presence in a typical textile park in

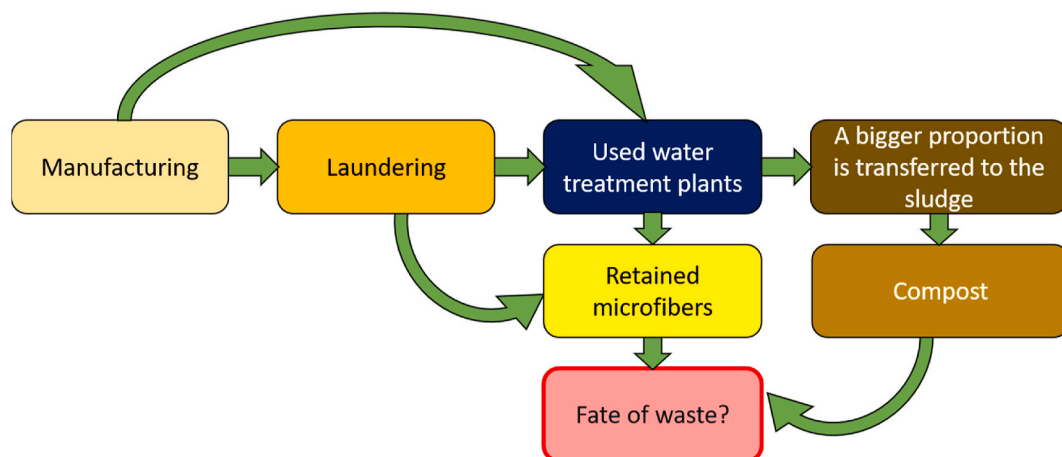


Fig. 1. Summary of the information included in this document.

Table 1

Summary of better practices in the textile industry published by [Mermaids \(2018\)](#).

Issue	Better practice	Main barriers
Melt spinning process	- Adjust to preserve fibers' mechanical properties. - Fiber fineness should be increased to decrease yarn tendency to form protruding MFs. - Fiber irregularities increase the friction between fibers and avoid the release of MFs from the yarn.	- Lower temperatures will increase the production time. - Yarn modification will alter its characteristics.
Drawing, stretching, texturing, intermingling and drying	- Adjust to preserve a good fiber tensile strength. The higher the tensile strength of the yarn the lower the probability of releasing MFs during washing. - Length of the fibers should not be too low.	- Process modification depends on the client's requirements.
Spinning (yarn)	- Continuous fibers detach less MFs than discontinuous or staple. - Plied yarns detach fewer MFs than single yarns. - High twist yarns detach fewer MFs than yarns with a lower twist. - The lower the linear density of the yarn (yarn count) the lower the number of fibers per cross-section the lower the release MFs.	- Yarn specification depends on the client's requirements.
Dyeing	- Avoid garment dyeing as it releases more MFs than yarn dyeing.	- The solution is to avoid or reduce dyeing.
Knitting and weaving	- The yarn carrier velocity of the knitting may be reduced to decrease the damage of the fiber. - The quantity or the nature of the sizing agent in the weaving process could be optimized and the velocity of the weft transporter could be reduced. - High-density fabrics have a tighter structure than lower ones, reducing the release of MFs. - Plain weave fabrics detach fewer MFs than twill weave ones.	- Increase of the production time.
Mechanical finishing	- The condition in the napping process may be optimized to reduce the mechanical stress on the fabric and its weakening. - The cut fibers should be recollected and managed in the factory. - Singeing mechanical finishing avoids the MFs formation on the fabric surface.	- Client requirements. - Difficulties to adjust the equipment.
Finishing	- Finishing agents capable of protecting the fabric surface can be used in this process.	- The agent must be compatible with the other finishing treatments.
Manufacturing	- A preliminary washing of the textile article can be	- Textile manufacturing mills may not have the

Table 1 (continued)

Issue	Better practice	Main barriers
	- done before selling it to remove the MFs generated during textile previous processes.	- facilities for industrial washing and wastewater treatment. - Current water treatments might not be able of collecting the smallest MFs.

China. They found that these installations can release as much as 54×10^3 MFs/L in printing and dyeing wastewaters. However, The MFs' generation was seen to significantly vary between mills. Hence, the authors discerned between the treatments that are applied in each textile plant, supporting the idea that the release of MFs is crucially influenced by the raw fabric materials and the conditions and chemical products applied in each industrial process. In this sense, the mill with the most concentration of MFs worked with rayon as raw material, which is usually subjected to heavy treatments like high temperatures and pressures. It has to be mentioned that, under the definition of the ECHA, rayon MFs are also considered as MPs. This means that the MPs' pollution could be underestimated because usually many authors do not include rayon MFs as a type of MPs. In addition, textile printing and dyeing might be a more significant source of MFs than domestic washing, which is on the order of half a million tons per year ([Belzagui et al., 2020](#)). According to this work, the studied mills have their own water treatment plants (WTPs), in which the efficiencies went from 85% to 99% of MFs' removal. The final effluents of these WTPs are subsequently treated in a centralized facility, from where they estimated a MFs' release of 430 billion particles per day.

On the other hand, producing and manufacturing textiles from natural rather than man-made or synthetic fibers has also been mentioned as a possible alternative to reduce the MFs' pollution. This statement is, at least, debatable, as nowadays most of the cotton business is based on a highly pollutant and environmentally unsustainable production ([Allary, 2021](#); [Garcia et al., 2019](#); [Maraseni et al., 2010](#); [Rukhaya et al., 2021](#)). For instance, cotton production uses 2.5% of world's farmlands but consumes 25% of world's insecticides and 80% of total water usage in the textile sector ([Garcia et al., 2019](#)). Hence, this claim can only be considered as an option in particular cases where it can be proved, for instance, by making a life cycle analysis (LCA) that the declaration is true. Regarding only the MFs' detachment, [Salvador et al. \(2020\)](#) reported that cotton articles shed more MFs than synthetic ones. However, these findings must be treated carefully to avoid misleading and generalized conclusions. In other words, there is a lack of inter-laboratory repeatability to ensure that the outcomes are accurate.

With regards to mixtures of materials, i.e., blends of synthetic and natural polymers, a reduction of MFs detachment might be achieved. In this sense, [Napper and Thompson \(2016\)](#) reported that, independently of the washing treatment, polyester - cotton (65/35%) blend fabrics detached fewer MFs than garments made only of polyester or acrylic. Conversely, [Zambrano et al. \(2019\)](#) found that polyester-cotton (50/50%) blends, as well as cotton and rayon ones, detached more MFs than polyester fabrics. This might again suggest that the fabrics' specific manufacturing process and the proportion of materials play an important role in the MFs shedding rate. Besides, polyester, cotton, and their blends receive different dying treatments that could be an important parameter to be furtherly examined. However, mixed materials can hamper the possible subsequent recycling procedure ([Herweyers et al., 2020](#)). Hence, it is of special interest to find the balance between the optimal blending proportion and the possibility of future uses or recycling of the discarded textile articles.

Alternatives can also consider the production of more resistant and/or higher quality fabrics. In this line, the quality of the products is an important parameter that should be considered. The current "fast-fashion" business model induces a decrease in social and environmental

conditions and produces low-quality garments that are more prone to detach MFs (Peters et al., 2021; Zamani et al., 2017). Hence, manufacturers and consumers can play a key role by investing in garments that are manufactured to last more and detach less MFs (Patagonia, 2018). Additives can be also considered, for instance, a 90% reduction of the MFs release was obtained by applying pectin, poly-lactic acid, and polybutylene succinate onto polyamide fibers (De Falco et al., 2019; De Falco et al., 2018a). Nonetheless, further investigation is required to develop additives able to avoid or reduce the MFs' detachment from chemically more stable materials (i.e., polyester) but without using toxic compounds as, e.g., methanol. For instance, Martel et al. (2002) explored the finishing of polyester fabrics with cyclodextrins and polycarboxylic acids. The study had no relation with MFs, however, they found an easy way with non-toxic chemicals to physically but permanently adhere a coating onto polyester fabrics. This type of studies can be assessed to apply them with MFs' reduction purposes.

Many works have studied the coating or surface functionalization of fabrics. These investigations can also be furtherly considered as alternatives to reduce the MFs pollution and stop them from reaching aquatic environments (e.g., Carosio et al., 2014; Chen et al., 2012; Cireli et al., 2007; Glampedaki et al., 2012; Rojas and Azevedo, 2011; Trad et al., 2018). Besides, it is equally important to make the additive or process sustainable. Using toxic or hazardous compounds or highly energetic demanding procedures in the production line could develop worst impacts than the benefits from the reduction of MFs.

3. Household and industrial washing machines, tumble dryers and additives

To the best of our knowledge, Browne et al. (2011) were the first to suggest that the MFs' contamination is related to the laundering of textile articles. In their publication, they noted that receiving water points of sewage effluents contained a higher abundance of MPs when compared to other reference sites. Besides, they found that the proportion of polymeric materials found in the sewage effluents were similar to the MPs contaminating shore sediments and disposal sites. Subsequent investigations verified and quantified the detachment of MFs from laundering processes, which is millions of MFs per cycle. In a first estimation, it was calculated that about 0.5 million tons of MFs reach the oceans every year (Boucher and Friot, 2017). However, it was later noticed that the estimation used a parameter that was 100–300 times larger than a more proper value. In particular, the linear weight of the MFs was considered 300 dtex (1 dtex = 1 g of fiber per 10 000 m), whereas the MFs will have a linear weight between 1 and 3 dtex. In this sense, Belzagui et al. (2020) re-estimated the MFs' flow by applying a methodology with new parameters and they obtained 0.3 annual million tons of MFs' stream to aquatic environments. However, if the ECHA definition proposal for MFs is accepted (length 0.003 mm–15 mm and a length/diameter ratio >3), only Pirc et al. (2016) have evaluated MFs sized above 5 mm. In this case, and adding that rayon MFs can also be considered as MFs, the amount of MFs detached per laundering reported by most of the published articles will be probably underestimated.

Common clothes and housing linens MFs' emissions were studied by Galvão et al. (2020). They assessed the detachment from a mix of 205 daily used textiles. They found a MFs release of approximately 3 million MFs per kg of washed article. This is in line with all previous studies. Besides, they reported that more than 90% of the MFs have a length <500 µm, with half of them being <100 µm. The washing temperature and time of the cycles were assessed by Cotton et al. (2020). They compared a "standard" washing cycle (40 °C and 85 min) with a "cold-quick" program (25 °C and 30 min). As expected, they found that higher temperatures and longer times significantly increased the MFs' detachment, reduced the garment longevity and accelerated the color loss, which is in line with the work of De Falco et al. (2018b). However, an interesting finding was that when the washing time is increased to a certain point, the fabric abrasion (and the MFs' release) become stable.

The estimated time stabilizing point was found at 35 min. However, no mechanism explanation was found to this behavior (Bao et al., 2017).

Furtherly, Dalla et al. (2020) evaluated the MFs' detachment from 100% polyester knitted fabrics with different operational washing conditions. They reported that the operative conditions (program time, temperature, speed of centrifugation, number of drum inversions, etc.) have a direct impact on the "stressing" or friction action on the garments. In this sense, the minor stress applied to fabrics was seen during "delicate" program, which were found to decrease the MFs' release by 16% when compared to the "cotton" program. However, it must be furtherly tested to understand if this is a generalized situation or a particular observation achieved in the washer used. If it happens to be a generalized condition, the appliance industry could apply these data to improve their programs to achieve the reduction of the generation of MFs.

Regarding the type of washers, Hartline et al. (2016) reported that top-loading washing machines detach more MFs than front-loading ones. As hypothesized by the authors, this may be a consequence of the central agitator of the top-loading models. The same study indicates that highly populated regions as Asia and North America have a high percentage of top-loading washers (90%). Hence, it could partially explain the usually big concentration of MFs found in the environment. To estimate what could happen if those countries have more front-than top-loading washers, Belzagui et al. (2020) studied an inverted scenario (10% of top-loading). This modification resulted in a global MFs' reduction of approximately 30%. This change implies a direct benefit to the aquatic systems but also an indirect one to the whole environment. However, as mentioned before, further investigation is needed to determine if this is a generalized situation or it only applies for the washers used in Hartline et al. experiments (Hartline et al., 2016). In this sense, improving the designs of washers to cause less friction to the garments or the inclusion of built-in MFs' filters can be also considered as feasible alternatives. By doing so, white goods manufacturers can include in their catalogues a qualification class regarding MFs' emissions of their machines. In this way, consumers will be able to consider this factor when acquiring a new washer.

Concerning the washing additives, there is still big uncertainty about their relevance in the detachment of MFs. For instance, Zambrano et al. (2019) encountered that the usage of detergent and higher temperature conditions showed a significant increase in the detachment rate for cotton. In this line, De Falco et al. (2018b) reported that liquid and powder detergents increase the MFs' release. They also found that powder detergents might cause a higher release of MFs due to their inorganic and water-insoluble compounds (like zeolite) that can produce friction to the textiles, and also due to the higher pH of these detergents. However, Napper and Thompson (2016) reported that no clear trend was found with the presence of detergent and conditioner, although fewer MFs were "occasionally" found when no-detergent or bio-detergent were applied. Moreover, Salvador et al. (2020) reported that the use of liquid detergent reduced the MFs' detachment when compared with no detergent for synthetic fiber garments (polyester, acrylic, and polyamide) but not for cotton ones. In this sense, making an extended assessment of the effects of the additives will benefit the formulation of new cleaning agents able to protect the garments and reduce their MFs' detachment. The findings reported by these articles are summarized in Table 2.

Although most methods shown in Table 2 exhibit similar results, there is an urgent need for a standardized analytical method (Henry et al., 2019). In this sense, some textile and interdisciplinary coalitions are working towards a unique analytical method. Indeed, in a recent publication from Tiffin et al. (2021), they proposed and published the validation of a method applying a Gyrowash machine. Besides, an inter-laboratory study was executed and they found that the method presented good replicability. However, it is important to mention that Zambrano et al. (2019) found that accelerated laboratory launderings, like those made in a Gyrowash, might release more MFs than household

Table 2
Published works regarding the MFs' detachment in industrial and household laundering.

Work	Analytical method/Comments	Results
Browne et al. (2011)	- No clear information on the methodology used. - Conservative estimations reported.	130–280 MFs/L per garment >1900 MFs/garment
Napper and Thompson (2016)	- Indirect method ^(a) . - Fibers with a mean length > 5 mm considered ^(b) .	140 000–730 000 MFs/6 kg of washed garments 500 000 MFs/mg
Pirc et al. (2016)	- Indirect method ^(a) . - Filters of 200 µm. - Mean length considered > 5 mm.	135 000 MFs/6 kg of washed garments
Hartline et al. (2016)	- Indirect method ^(a) .	29–431 mg of MFs/garment-washed (front-load) 1471–2121 mg of MFs/garment washed (top-load)
Carney Almroth et al., 2018	- Gyrowash one bath 815/8 according to modified SS-EN ISO 105-C06. - Commercial liquid detergent.	7360 MFs/(m ² L) for PES fleece 87 MFs/(m ² L) for knitted PES
Salvador (2017)	- Used Napper and Thompson (2016) and Pirc et al. (2016) methods.	184 000–250 000 MFs/garment
De Falco et al. (2018b)	- Linitest apparatus used to simulate washing machine.	6 000 000–17 700 000 MFs/5 kg of washed garments 0.43–1.27 g of MFs
Belzagui et al. (2019)	- Direct quantification. - Commercial front-load washer. - Quantification of the MFs by visual counting. - Polyamide 20 µm filter.	1 000 000–6 500 000 MFs for 6 kg of washed garments 30 000–230 000 MFs/garment 2–29 mg of MFs/garment washed 1200–33 000 MFs/(m ² L)
Zambrano et al. (2019)	- AATCC standard SDL Atlas Launder-Ometer. - Whatman glass 1.2 µm filter. - HiRes Fiber Quality Analyzer (FQA) for MFs' quantification. - 25°C and 44°C. - AATCC 135–2015 with Washing machine - Nylon 20 µm and then Whatman glass 1.2 µm filter.	<i>Mass lost during accelerated laundering:</i> 0.2–4.0 mg MFs/g (cellulose-based fabrics) 0.1–1.0 mg MFs/g (polyester fabric) 0.1–1.0 mg MFs/g (polyester fabric)
Salvador et al. (2020)	- Commercial top washer used in the experiments. - Subsequent steel sieves 500 µm and 65 µm. - Indirect method to estimate the weight of the MFs (applying the diameter, linear weight, and density).	% of lost garment weight from 10 cycles: 0.16 to 0.20 (100% cotton) 0.07 to 0.13 (100% acrylic) 0.03 to 0.06 (100% polyester) 0.04 to 0.10 (100% polyamide)
Galvão et al. (2020)	- Clothing and housing linens from a household of 4 people. - Commercial front-loading washer. - Aliquots of 20 mL were filtered through 12 µm nitrocellulose membrane filters - MFs were visually counted in one quadrant of the filter and extrapolated.	18 000 000 MFs for 6 kg of washed garments
Dalla et al. (2020)	- 100% PES knitted fabrics. - Pre-washing before experiments. - Commercial washer, detergent, and softener used. - Effluent was filtered through 40 µm pore. - Different programs tested.	36.60 mg/kg (cotton program) 32.51 mg/kg (delicate/silk program with liquid stain remover) 33.86 mg/kg (delicate/silk program.)

(a) Indirect method: the quantification is estimated from the weight, length, and/or density of the MFs.

(b) According to the ECHA, fibers with a length <15 mm and a maximum diameter of 5 mm should be considered as MFs.

ones. This could indicate that these types of tests might only be applicable when relative detachment rates are the objective of the assessment. In other words, the absolute outcomes of MFs' detachment might be evaluated in real conditions (conventional washers, textile articles and additives).

It should also be considered that each methodology has limitations or disadvantages. For instance, Belzagui et al.'s (2019) method and similar procedures are time-demanding. Besides, visual counting might be preferably used when small amounts of garments with strong colors are tested. On the other hand, methods such as those proposed by Salvador et al. (2020) might not be able to discern between MFs and longer fibers, or could erroneously include the mass of the detergent or other impurities in the weighting process of the MFs. To that effect, a method that considers different situations, types of samples or requirements will probably be needed, for example, one for relative and the other for absolute amount of released MFs.

On the subject of dryers, O'Brien et al. (2020) studied the airborne emission of MFs from a domestic vented dryer. They found that the lint emission corresponded to approximately 0.012% of the garment mass dried. Kapp and Miller (2020) also studied the MFs emission from dryers. They collected the MFs vented to the surroundings of the experiment site and provided the results in different units for further comparability. In this sense, they reported averages from 35 mg to 70 mg of lint from three consecutive dry cycles. In addition (Mahbub and Shams, 2022), reported that the MFs release was promoted by longer drying times. As can be seen, these studies confirmed that MFs can escape the lint trap of the dryers and therefore should be considered as a source of MPs to the environment. However, further investigation is needed to assess the detachment rate from these machines. In this sense, clothes tumble dryers can also be modified to be more efficient regarding the MFs' generation. For instance, TeGrotenhuis et al. (2017) proposed a hybrid heat pump dryer that could achieve savings in both energy and drying cycles, providing an indirect reduction of the MFs generated.

3.1. Devices to reduce the MFs in the effluents of the laundering equipment

There are commercially available devices to reduce the release of MFs from washers. These might work by reducing the MFs either inside the washing machine [in-drum devices such as GuppyFriend (2021) or Cora Ball (2021)], or at the effluent [external filters as Lint LUV-R (2021) or PlanetCare (2021)]. These devices work differently when reducing the concentration of MFs in the washers' effluent. GuppyFriend and analogous systems work basically to protect the garments from the mechanical stress generated in the washing process. Hence, it is expected that they will reduce the generation of MFs. In contrast, external filters and the Cora Ball act by retaining the already generated MFs. It must be noticed that these devices might be complementary between them to achieve a higher reduction of the MFs.

According to McIlwraith et al. (2019), these technologies can accomplish a MFs' reduction in the washer effluent from 26% (Cora Ball) to 87% (Lint LUV-R filter). Additionally, no significant difference for the MFs length when using the Cora Ball was reported, suggesting that this device captures MFs in a wide range of sizes. Meanwhile, Napper et al. (2020) reported that the external device XFiltra (2021) was the most successful in retaining MFs, followed by the in-drum GuppyFriend. For the XFiltra, two main explanations were given: a finest mesh pore in contrast with other similar devices (60 µm vs > 175 µm), and the use of an integrated electrical pump to facilitate the flow through the mesh. On the other hand, PlanetCare filter sustains more than 90% of MFs'

retention efficiency for their product, which is not consistent with the 25% retention reported by Napper et al. (2020). The results of these studies are summarized in Fig. 2.

As can be seen in Fig. 2, different results are reported for the same devices. Hence, further experimental inter-laboratory replications are required. The introduction into the market of these options seems feasible in the short-to medium-term period. Nevertheless, the final disposal of the retained MFs is an urgent concern not addressed yet. For instance, as commented by Napper et al. (2020), once the filters are cleaned by collecting the MFs, these can be “thrown into the everyday household waste”. In this sense, depending on the final disposition given to that specific waste, the MFs could also finish in water environments, turning the devices just into a mere MFs’ “by-pass”.

The usage or final disposition of the MFs has been poorly studied. For instance, Yousef et al. (2021) have proposed the use of lint from household dryers as a source of renewable energy through pyrolysis. They found that lint can produce an activation energy higher than the textile waste as a result of its high purity, chemical composition, and uniform size, which facilitates the conversion process. However, it must be noticed that many of the lint materials, like polyester, will not have a renewable origin. In this sense, this fossil-based waste should not be considered as a renewable energy source. Besides, there is a logistic issue that needs further consideration to transport this waste. A good idea would be to persuade the users who have a MFs’ retention system to take them to a MFs’ “disposal” site, from where these could be collected and submitted to a future final disposal treatment facility. In this sense, we can make an analogy with gases’ emissions. There are diffuse emissions (cars – household laundering), which are harder to treat, and point sources which are easier (fabrics – textile industries). In this line, a good alternative could be to reuse them when these are collected in industries (point sources) and to immobilize them in treating facilities when these are retained in household laundings (diffuse sources).

Currently, textile articles are washed even if they are not dirty, which implicates a futile use of water and energy resources (Stawreberg and Wikström, 2011). Hence, an effective and obvious method to reduce the MFs’ shedding is to wash them only when it’s required. For instance, in a

hypothetical situation simulated in the work of Belzagui et al. (2020), a reduction of 30% on the generation of MFs can be achieved by reducing the laundering water consumption in the regions with a higher consumption rate (which is 19 m³/washer per year). In other words, doing shorter washing cycles or washing less generates fewer MFs. This measure can be accomplished by instructing consumers to use adequate laundry programs and/or more efficient washers. This strategy can be implemented in the short-to medium-term to reduce the MFs’ generation. Consequently, this will decrease the emission of MFs to the whole environment, affecting especially to the usage and contamination of water ecosystems.

It must be noticed that “washing less” is very subjective. For this reason, there are some guides to provide sufficient knowledge to consumers about when and how to wash their clothes. For instance, reported “better practices” are: filling up the washing machine, using liquid detergents, selecting colder and quicker laundry settings, among others (Cotton et al., 2020; De Falco et al., 2018a; Mermaids, 2017; Plastic Pollution Coalition, 2017). This could be strengthened with textile campaigns indicating how many “normal” uses are appropriate for a determined garment before washing. In this way, the population aware about the reduction of MFs generation must be continuously reinforced. Social networks are strong tools that are constantly making publications on this topic (GESAMP, 2015b; SAPEA, 2019; Wagner and Lambert, 2018); however, environmental education should be guaranteed from the early stages of our education, and MFs should be a part of this education.

Regarding the mass or quantity of garments washed per cycle, in a quick experiment made in our laboratory to measure the MFs’ detachment, polyester fabrics were independently submitted to one washing cycle at equal conditions (Fagor Innovation F-2180 washer, 40 °C, 1000 RPM, 57 min, and 59 L of effluent). The textiles used were identical fabrics but differing in their weight and dimension; two weighted 0.13 kg, while the other pair weighted 3.00 kg (4 samples in total). Each sample was washed with liquid detergent. In this case, the contrast was from a washer occupation of 25%–70% in volume and from 20% to 50% in weight. The relation between the mass of garments is about 23 (3.00

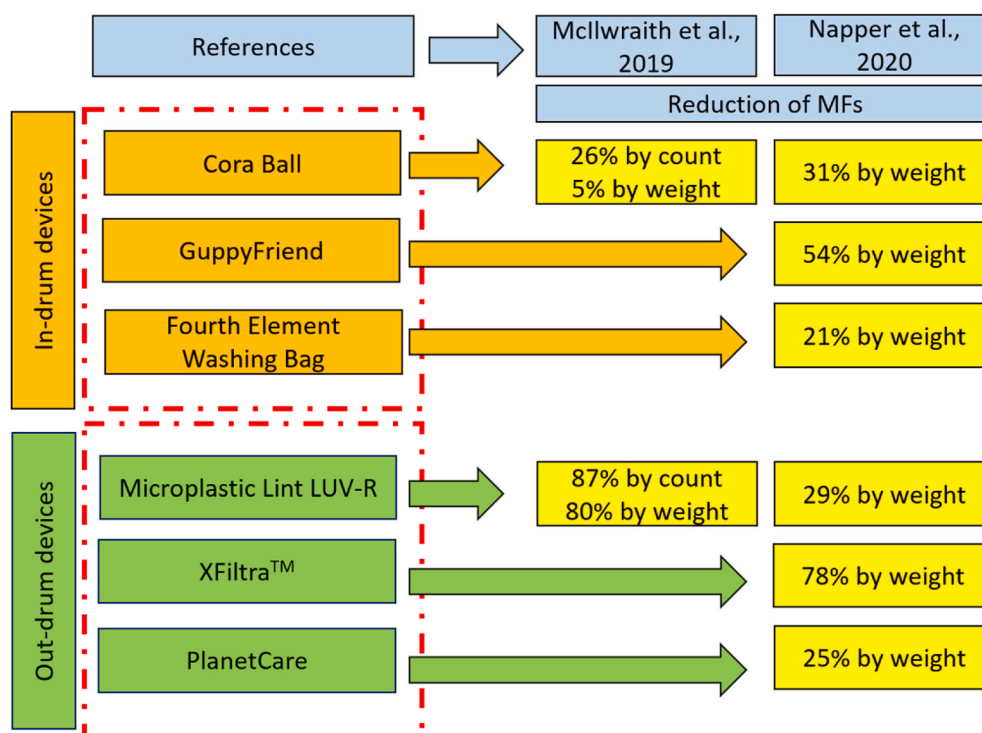


Fig. 2. Microfibers’ reduction at the effluents of the washing machines.

kg/0.13 kg), while the relation between the mass of the MFs detached was seen to be approximately 5 (0.025 g/0.005 g). The reduction of the MFs could be a consequence of a decrease in the friction between the textile articles and the washer drum. Besides, this quick experiment verified what is said in the “Mermaids good practice guide” (Mermaids, 2017): filling up the washing machine considerably reduces the MFs detachment.

4. Municipal, industrial, and drinking water treatment plants

This section discusses the MFs and MPs found in water treatment plants (WTPs) and water products. The addressed perspective is to inform about the concentrations typically encountered in the effluents, the effectiveness of the different technologies for the MPs' removal, and the effects of the MPs on the treatments.

4.1. Drinking water treatment plants

It has been published that the consumption of bottled and tap water is a source of ingestion of MFs (Ossmann, 2021). The average concentrations reported were 94 MPs/L for bottles and 32 MPs/L for tap water (Eerkes-Medrano et al., 2019; Koelmans et al., 2019; Mintenig et al., 2019; Schymanski et al., 2018). Cox et al. (2019) estimated that the annual MPs' intake could be 52×10^3 items. However, it may raise to 90×10^3 MPs if only bottled water is consumed. Kosuth et al. (2018) found that most of the particles were MFs (98%). Regarding the size, Ossmann et al. (2018) reported that over 90% of the MPs found in bottle water corresponded to particles smaller than 5 μm , which are smaller than the mean 960 μm reported by Kosuth et al. (2018) for tap water. Hence, drinking WTPs are not completely effective in eliminating MPs, whether as for uncompleted retention or as for an “in-situ” generation.

In this line, Wang et al. (2020) reported that the overall MPs' removal from different technologies in drinking WTPs was from 82% to 89%. On the other hand, Zhang et al. (2020a, 2020b) found a low retention of micro- and nanoparticles (<2%), with an increase of 16% when applying a coagulant aid. Works from Ma et al. (2019a, 2019b) also reported low retention efficiencies (from 1% to 8%) at coagulants' conventional dosages. Hence, there is a need to mend these contradictions. Regarding the ultrafiltration process, it has been reported as the most effective process for MFs' removal (Ma et al., 2019a; Zhang et al., 2020b). Sand filtration was also tested but it was not considered to play a primary role in removing MPs. A key finding was that drinking WTPs can also act as a source of MPs. In this specific case, greater concentrations of polyacrylamide (PAM) were detected in the effluent compared to the raw water. PAM is used as a component of the coagulant used in the process. In this sense, more investigation is needed to evaluate if the presence of MPs and MFs in drinking water is due to the physical degradation of the plastic bottles or the water treatment itself (Ossmann, 2021).

4.2. Used water treatment plants

In municipal or industrial used-WTPs, a proportion of the incoming MFs will be transferred into the sludge throughout the consecutive treatments. It is important to notice that WTPs should not be considered as a source of MPs and MFs but a pathway where they can be removed from the liquid stream. Globally, the percentage of municipal used-waters that are subjected to any kind of management is approximately 20% (ONU, 2017; Pham and Kuy, 2013). Populations connected to urban WTPs are markedly varied across the countries. For instance, South American and Asian countries treat around 20% of their municipal waters, while Central European countries have achieved a 97% of treatment coverage (EEA, 2017; Mara, 2004). WTPs are considered significant pathways for all types of MPs to aquatic and soil environments (Raju et al., 2020; Rolsky et al., 2020). Yet, as mentioned before, fiber-shaped ones coming from the laundering of textile articles are

within the most encountered types in these streams.

Regarding the material, polyester MFs usually surpass other types of MPs (Browne et al., 2011; Lares et al., 2018; Magnusson et al., 2016; Murphy et al., 2016; Sun et al., 2019). Currently, polyester is the top synthetic material used by the textile industry. Besides, as pointed out by a series of publications, regardless of the material, each textile article detaches thousands to millions of MFs in every domestic washing cycle (Belzagui et al., 2019; De Falco et al., 2018b; Folkö, 2015; Hartline et al., 2016; Pirc et al., 2016). For instance, Alavian Petroody et al. (2020) reported that most of the MPs found in a WTP were in the form of MFs, from which polyester was the most abundant, followed by polyamide and acrylic fibers. This makes sense, as an important part of the MPs entering a WTP will come from the household washers' effluents. Following the current data, it has been estimated that 0.48 million tons of MFs are globally generated in domestic washers, from where 0.20 million tons might be retained in WTPs' sludge and 0.28 million tons might reach aquatic environments (Belzagui et al., 2020).

In primary treatments, raw water usually passes through a set of screenings. If fine screens are applied (2.5–10 mm), MPs >2.5 mm can be removed from the water but not completely because of the morphology of these particles (big length to diameter ratio). Next, grit removal, flotation, and primary settlement could also retain MPs with different densities than water (Sun et al., 2019). Primary treatments are effective to retain MFs as these might be adsorbed, aggregated, and entrapped in flocculating particles and separated by sedimentation (Sun et al., 2019; Wei et al., 2019c). An important removal of the MFs can happen in this stage. In some cases, MPs' removal was found to be even greater than 90% (Lares et al., 2018). Secondary treatments, as activated sludge, are basically biological processes that degrade organic pollutants. Here, MPs can be retained by the extracellular polymer substances secreted by microorganisms and furtherly removed with the generated sludge; i.e., MPs are transferred to the sludge (Sun et al., 2019). In general, as reported in Bakaraki Turan et al. (2021), most of the MPs' and MFs' removal is made in the primary and secondary treatments. Finally, advanced treatments as coagulation and filtration, ultrafiltration, or membrane bioreactors can remove part of the remaining and lower-sized MPs. As can be seen in Fig. 3, the retention and transfer rates might depend on the treatment applied. From different studies (Alavian Petroody et al., 2020; Blair et al., 2019; Carr et al., 2016; Franco et al., 2020; Gasperi et al., 2015; Gies et al., 2018; Gündoğdu et al., 2018; Lares et al., 2018; Leslie et al., 2017; Magni et al., 2019; Magnusson and Norén, 2014; Michielssen et al., 2016; Mintenig et al., 2017; Murphy et al., 2016; Talvitie et al., 2015; Talvitie et al., 2017a, 2017b; Zhang et al., 2021; Zhou et al., 2020), it can be stated that the MPs' and MFs' transfer efficiency will be between 76% and 98%.

For instance, Lares et al. (2018) reported that 99% of the MPs were removed in the primary treatment; afterward, a slight increase of the MPs' concentration (0.6–1 MPs/L) was found in the effluent of the activated sludge. This behavior might be a consequence of the sampling procedure, which didn't consider the different flows (turbulent and stable) that are found in each effluent. Hence, superficial sampling might retain more buoyant MPs when having a stable flux. MPs with a density higher than water will be mostly retained in primary and secondary treatments, whilst advanced treatments will eliminate floating particles from the final water effluent (Nizzetto et al., 2016). In this line, Bayo et al. (2020) found that particulate shapes were more prone (95%) to be retained than fiber ones (55%) in membrane bioreactors (MBR) and rapid sand filtration (RSF). It should be noticed that MFs might have an easier pass through RSFs as a consequence of their longitudinal shape and small diameters ($\sim 10 \mu\text{m}$) (Hamidian et al., 2021). Edo et al. (2020) reported that sizes between 25 and 104 μm were predominant in the effluents of primary and secondary treatments, strengthening the conclusion that smaller and fiber-shaped MPs are more prone to be found in the effluents of WTPs (Li et al., 2020b; Raju et al., 2020).

Despite having a relatively high retention efficiency, these facilities treat millions of liters every day, releasing high amounts of MFs (Sun

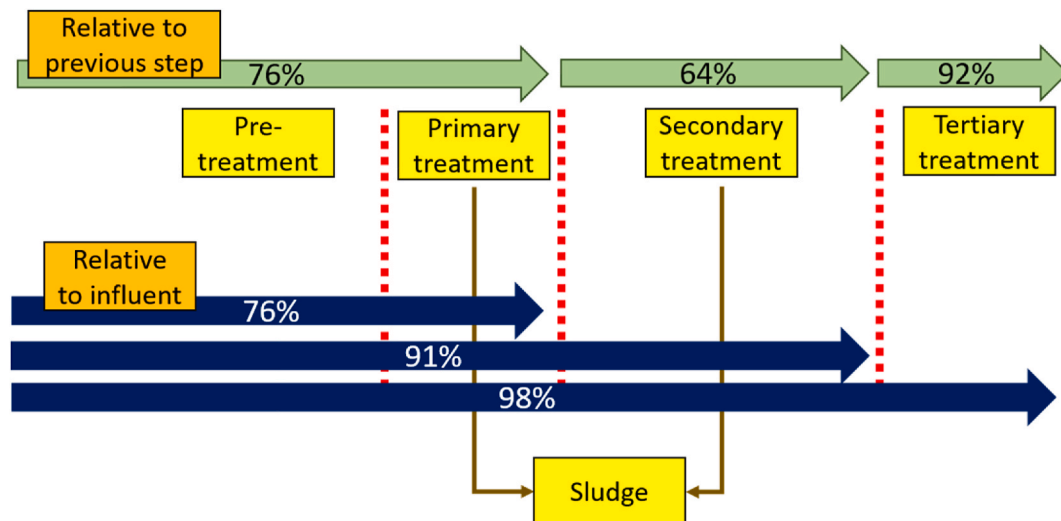


Fig. 3. Percentage of retained microplastics and microfibers in a used-water treatment plant.

et al., 2019). Also, as previously explained, the proportion of treated waters is still very low across the world, and it must be noticed that other sources of textile MFs (as garments' hand-washing) will be still left aside from these treatments. Equally important is that these particles can still enter the environment via the final disposal of the sludge, as conventional treatments don't remove sludge-based MPs (Z. Chen et al., 2020). The reported abundance of MPs in the sludge varies from 1.5×10^3 to 180×10^3 particles per kg of dry weight sludge (Edo et al., 2020; Lares et al., 2018; Li et al., 2020b). Hence, MFs might still be dumped into the environment if the sludge is used as, e.g., an agricultural fertilizer (Bayo et al., 2016; Corradini et al., 2019; Li et al., 2018; Mahon et al., 2017).

The existence of WTPs was estimated to play a relevant role to transfer MFs from the liquid stream into the sludge. According to Belzagui et al. (2020), a global 30% reduction of MFs reaching aquatic environments can be achieved by increasing regions with a low percentage (<50%) of treated water to 60%. However, as pointed out in that study, further investigation is required to develop treatments for the MPs and MFs retained in the sludge. Yet, it must be noticed that installing only primary treatments in places without WTPs could help to remove an important proportion of the incoming MPs and MFs from the liquid stream.

As previously indicated, the amount of MFs annually retained in the sludge of WTPs is about 0.20 million tons (Belzagui et al., 2020). Also, a publication estimated that the yearly amount of MPs entering agricultural lands from sludge might be between 63×10^3 to 430×10^3 and 44×10^3 to 300×10^3 tons in Europe and North America, respectively (Nizzetto et al., 2016). In this latter study, other sources of MPs, as particles collected by sewers, were also considered in the estimation. Partial removal of these fibers can be achieved by sieving and sifting procedures, however, a complete separation will not be possible (Weithmann et al., 2018). A recent promising strategy to reduce the MPs' concentration from the sludge is the hyperthermophilic composting technology (hTC), which was demonstrated to reduce 45% of the MPs after 45 days of treatment at a full-scale trial (Z. Chen et al., 2020).

Regarding passive treatments, Sarkar et al. (2021) studied the MPs' and MFs' pollution in freshwater wetland systems used for wastewater treatment. They estimated a MPs' removal of 50%, which is far lower than most conventional primary treatments. Besides, an important aspect to consider is that they identified variable amounts of heavy metals (As, Cd, Cr, Cu, Ni, Pb, Zn), ranging from 2.03 µg of arsenic per gram for MPs to 191.01 µg of zinc per gram of MPs. Fishes in the wetland ponds were also contaminated with MPs and heavy metals. In this sense, they concluded that natural wetlands are facing the risk of MPs pollution, where MPs have gotten to the trophic state hindering and stressing

the wetland system. This article is in line with Li et al. (2020b), who reported an abundance of 5.5×10^3 MPs per m³ of water in a freshwater mangrove. However, as stated by Kumar et al. (2021), the number of MPs in wetlands will depend on various factors, such as the location, proximity to urban settlements, human interference, among others. The MPs' removal in natural wetlands could be very helpful for economically under-developed countries. In any event, it is evident that retaining the MFs before reaching WTPs will reduce their contamination to the environment, i.e., short-to medium-term alternatives as washers' filters can be a good alliance to reduce the flow of these pollutants either to water, soil or atmospheric ecosystems.

4.3. Effects of MPs and MFs on water treatment plants

It has been reported that MPs in biological treatment might reduce the abundance of the bacteria that is needed for the processes of nitrification, denitrification, among others. For instance, some MPs might inhibit the sludge anaerobic digestion in all its phases; hydrolysis, acidogenesis, acetogenesis, hydrogen, and methane production (Wei et al., 2019a, 2019b; Zhang et al., 2020a; Zhang and Chen, 2020). In some cases, depending on the polymer, the effects might have different origins. In this regard, polyethylene terephthalate (PET) and polyethylene (PE) induce the formation of reactive oxygen species (ROS, as OH[•] or H₂O₂). PET and Polyvinyl chloride (PVC) can release bisphenol-A (BPA) and di-n-butyl phthalate (DBP), respectively, both being toxic compounds (Wei et al., 2019a, 2019b, 2019c). Minimum concentrations for the effect were established at 10, 20, and 100 MP/g for PET, PVC, and PE, respectively, which are realistic concentrations found in sludge samples (Li et al., 2018; Wei et al., 2019c). On the other hand, polystyrene nanoparticles affected the microbial community structures by reducing the cumulative methane production by 15% (Fu et al., 2018). Regarding polyester MFs, Li et al. (2020a) found a reduction of methane production for several concentrations. Yet, the inhibition was lower than with other tested polymers. Also, Qin et al. (2020) tested with polyethersulfone MPs, and it was found that these particles slightly reduced the removal of ammonia nitrogen. In this sense, it is also important to have a better understanding of the effects that MPs and MFs will have on the biota of WTPs. For instance, if the potential damage is elevated, it will be necessary to remove them as much as possible before reaching WTPs.

5. Gaps, criteria, and phases of the solutions

As in every environmental issue, the solutions must fulfill a minimal

set of conditions to be considered practical and to provide a positive net effect on the environment. Some of these criteria are mentioned hereafter:

1. An effective solution should avoid producing secondary issues when tackling the main concern. There are many ways to produce secondary problems, for instance, high energy and resources requirements, the usage of toxic or hazardous substances, among others. For example, the application of toxic compounds in the process could imply the subsequent event of environmental contamination (e.g., the use of methanol for producing textile coatings). Or, including washable coatings to textile articles could imply an increase of the BOD at the effluent discharging areas. In this sense, it could be of particular interest to perform a Life Cycle Assessment (LCA) for each MFs' processing or retaining system.
2. Alternatives should also seek the possibility of closing the gap of the MFs. In this line, some devices have shown great effectiveness in retaining MFs in washing machines. However, the interrogative of the posterior treatment must be equally and urgently solved.
3. The treatments or alternatives must be scalable. This is particularly complex in the manufacturing process of textile articles and the treatment of municipal waters. To better illustrate this, it must be noticed that the manufacturing of textiles is produced at uncountable points around the globe. In the same line, municipal waters are still poorly treated in terms of the worldwide proportion, being approximately 80% of the waters discharged to the environment without any treatment. On the other hand, including devices in the washing machines could be easier in the short- and medium-terms.
4. In the case of devices intended for the users' application, these must be practical and easy to handle. The higher the device complexity the lower will be the implication of the users. This is furtherly explained in the work of Herweyers et al. (2020).
5. The economical parameter is also an important aspect to ponder. Unless policies or regulations (laws) to reduce this contamination are created, devices and/or processes should not exceed a "critical" economical point in order to promote the users' engagement.

In this line, Herweyers et al. (2020) assessed the consumers' perceptions and attitudes toward systems preventing MFs pollution. With this purpose, they determined the optimum requirement for these systems through a consumer survey. They found that the MFs' problem and peoples' washing behavior are underestimated. Besides, they constructed a minimal set of requirements from a user point of view, which are summarized next:

- i. The solution must be effective and preferably visually experienced by the users so they can recognize their positive performance towards a cleaner environment.
- ii. It must be durable and ensure long-term usage, i.e., disposable solutions should not be considered.
- iii. The usage or installation of the product should be easy.
- iv. People from all socio-economic levels should be able to acquire the product.
- v. In case of a cleaning requirement of the product, it should be fast and user-friendly. The achievable amount of cleaning periods was found to be at every 15–17 washing cycles with a cleaning duration of 10 min.

Also, further research is needed to investigate the possibilities to close the loop for the collected MFs. There are a lot of gaps still to be solved and to be investigated, as the MFs emitted to the air from the daily usage of the garments, or ways to immobilize or use the retained MFs. However, it must be noticed that there are environmental issues for all the life cycle of the garments. For instance, as reported by the Ellen MacArthur Foundation, less than 1% of the material used to manufacture garments is recycled into new garments (Ellen MacArthur

Foundation, 2017). In this sense, the MFs are one of the missing pieces to be solved in order to make the textile industry more sustainable, but further investigation is needed to develop new ideas or to improve already existing ones.

6. Conclusions

Many alternatives are available to reduce textile microfibers from reaching the environment. Some options are currently more viable in the short- and medium-term periods. The textile industry has the potential to drastically reduce the generation of microfibers by improving their processes or products. This could imply that downstream solutions might be dispensable or less severe. However, there are many small- and medium-sized textile industries around the globe, making this alternative feasible only in the long-term time. Also, there are currently some solutions for washers. These can reduce at least 30% of the microfibers' emissions from household laundry. Besides, new products as detergents or additives are being developed to reduce the generation of these particles.

On the other hand, water treatment plants can partially remove the microfibers from the liquid stream and retain them in the sludge. Depending on the technology applied, these facilities can remove up to 99% of the microfibers. Yet, the problem is still transferred to the sludge. In addition, installing these facilities is a long-term alternative. An important gap in every alternative is the final disposition or treatment of the microfibers. It is important to clarify that any solution must consider the whole process to certify that it is environmentally friendly and will not pollute more than the microfibers. Yet, it is very likely that the alternatives will be complementary between them, i.e., there will be no single solution for the microfiber pollution.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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